
OpenWAM

Glossary and Field Index

Every component and every field: meaning, unit, and a sensible default grouped by palette tab, with the complete results-magnitude catalogue

The reference companion to the *OpenWAM User Guide*

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How to read this glossary

This is the field reference behind the *OpenWAM User Guide*: one entry per component, grouped by the palette tab it lives on, listing each field with its unit, meaning, and a sensible default. Defaults are tagged **[std]** (a conventional value), **[case]** (depends on the model), or **[set]** (a specific value validated in a repository example). A dash (—) in the unit column means dimensionless. Notes below a table carry the traps worth knowing. Where a field only sets a

boundary or a file format, the meaning names the file it reads; the *OpenWAM Developer Guide* maps each component to its source file.

Two conventions hold throughout. First, every element with a volume has a **Composition** tab (fuel / burnt gas / air mass fractions in the simplified species model; O₂, CO₂, H₂O, HC, Soot, NO_x, CO, Fuel, N₂ in the extended model); it is not repeated in each entry. Second, a duct never attaches directly to a zero-dimensional element — it passes through a connection element (a valve, throttle, stator/rotor, or Node to Deposit), so a 0-D element with a duct on each side needs two of them.

1 Nodes tab

1.1 Ramification (junction)

A Y-split or merge of two or more pipes. It has no UI fields — it is pure topology. Its gas dynamics use the GJM junction model, which conserves mass and stagnation enthalpy across the branches.

1.2 Union of two pipes

The default pipe-to-pipe joint.

Field	Unit	Meaning	Default
Junction thickness	mm	wall thickness at the joint	[case]
Conductivity	W/m K	wall thermal conductivity	50 (steel) [std]

1.3 Laminar / Turbulent Pressure Loss

An in-line stagnation-pressure-loss connector between two ducts (one duct on each side). *Laminar* (node type 9) makes the loss linear in velocity, $\propto FK \cdot |U|$; *Turbulent* (node type 10) makes it quadratic, $\propto FK \cdot U^2$. Mass is conserved across it.

Field	Unit	Meaning	Default
Coefficient of pressure loss	—	the loss coefficient FK . Must be > 0 (the solver divides by $\sqrt{FK}$). Example values 0.5 (laminar) / 0.75 (turbulent) are physical order-unity minor-loss coefficients.	> 0 [std]

1.4 Extreme to Atmosphere (open end)

An open intake or exhaust boundary.

Field	Unit	Meaning	Default
Losses coefficient	—	flow-openness multiplier, not an additive loss: 1 = fully open, 0 = sealed; applied as $u = u_{\text{isentropic}} \cdot \text{coeff.}$	1.0 [std]

1.5 Extreme with constant P and T

A prescribed reservoir end, typically an exhaust discharge.

Field	Unit	Meaning	Default
Losses coefficient	—	as for the open end (openness, 1 = open)	1.0 [std]
Pressure	bar	reservoir pressure	1.013 [std]
Temperature	°C	reservoir temperature	ambient [case]
Exhaust model	—	no = on inflow the end takes the reservoir composition (correct for a fresh-air intake supply); yes = composition left untouched (use when the reservoir stands in for the engine exhaust, so back-flowed burnt gas is not overwritten)	per use [case]

1.6 Anechoic Extreme

A non-reflecting end for acoustic work. No fields.

1.7 Closed Extreme

A capped end. No fields.

1.8 Screw Compressor (supercharger)

A positive-displacement supercharger (node type 13) modelled as a prescribed-inflow boundary — not a map-based turbomachine. Exactly one duct connects (the discharge); the suction side is an implicit atmosphere reservoir at the entered inlet conditions. Three empirical polynomials in the discharge pressure P set the volumetric flow, the delivered temperature, and the (reported-only) shaft power.

Field	Unit	Meaning	Default
Inlet Pressure / Temperature	bar / °C	suction conditions; set the intake density for the volume-to-mass flow conversion only (they do not create an upstream pipe)	1.013 / ambient [std]
Flow coefficients C1, C2, C3	mixed	$Q = C_1 + C_2P + C_3(SR \cdot N)$; C_2 is normally negative (delivery droops with back-pressure)	[set]
Temperature coefficients D1, D2, D3	mixed	$T_{out} = D_1P^2 + D_2P + D_3$	[set]
Power coefficients E1–E6	mixed	reported shaft power; keep E_6 small (it sits inside an exponential of speed and can overflow)	[set]
Speed control	—	Proper (fixed rpm, uses Speed) / Engine (slaved to engine, uses Speed ratio)	Proper [case]
Speed / Speed ratio	rpm / —	fixed compressor speed (Proper) or compressor-to-engine ratio (Engine)	[case]

Note. The .WAM stores the speed controls first, then the inlet P/T, then the coefficients — a different order from the UI. Outputs: Power_Roots_Comp, Mass_Flow_Roots_Comp, Output_Pressure_Roots_Comp.

1.9 Pressure Pulse

A wave-generator boundary (node type 5) that plays a pressure/entropy table against the pipe end; it fully prescribes the boundary and re-reflects returning waves (it is not anechoic).

Field	Unit	Meaning	Default
Time / Pressure / Entropy table	s, bar (rel.), —	one row per time point, entered directly or from a .DAT. Pressure is relative to the pipe's initial pressure, so a 0 entry launches no wave. Entropy level 1.0 = gas at reference temperature. First row must be $t = 0$; the table repeats cyclically (last row's time = period).	[case]

1.10 Static Pressure Inlet

A time-varying reservoir boundary. For inflow it prescribes static pressure and temperature; if flow reverses it acts as a constant-pressure open end.

Field	Unit	Meaning	Default
Time / Pressure / Temperature table	s, see note, °C	same .DAT container and cyclic repetition as the Pressure Pulse.	[case]

Note. Verify the pressure column on first export: the solver reads it as an absolute ratio, so if the UI writes a raw relative value a 0 entry means vacuum. Save a known value and read it back in the .WAM.

1.11 External Extreme Calculation — non-functional

The MATLAB co-simulation end. **Disabled in this build; do not use it** (User Guide, control chapter).

1.12 External Connection

A one-dimensional external (CFD) coupling; a single integer connection id. Confirm it is wired before relying on it.

1.13 Indefinite Node

A placeholder (“?” icon). Unused — do not place it.

2 Connections tab

2.1 Valve (poppet)

A four-stroke intake or exhaust valve.

Field	Unit	Meaning	Default
Valve diameter	mm	reference diameter	[case]
Valve opening	degree	opening angle in the engine 0–720° frame (0 = firing TDC, 360 = overlap TDC); closing follows from the lift-table length	[case]
Speed control	—	engine (reads the cylinder angle directly) / self control	engine [std]

Field	Unit	Meaning	Default
Lift interval / Angle step	mm / deg	row spacing of the flow and lift tables	[case]
Flow table	—	Cd vs lift, saved as .FLU (count, then rows of inflow Cd, outflow Cd, swirl)	[case]
Lift table	—	lift vs angle, saved as .LEV (count, then one lift in mm per line)	[case]

2.2 Throttle Valve (Mariposa)

A butterfly throttle between a duct and a 0-D element.

Field	Unit	Meaning	Default
Reference diameter	mm	throttle bore	[case]
Initial lift	—	starting opening fraction	[case]
Cd-vs-lift table	—	at least two rows (lift, inflow Cd, outflow Cd)	[case]

Note. An empty Cd table crashes the run (now guarded with a clearer error). The throttle is a connection to a 0-D element, so a bypass is *duct* → *throttle* → *chamber* → *Node to Deposit* → *duct*.

2.3 Node to Deposit

The connector wherever a pipe meets a 0-D volume (Figure 1).

Field	Unit	Meaning	Default
Kinetic-energy recovery	—	yes/no; selects which coefficient the next field is	no [std]
Discharge (or recovery) coefficient, duct → plenum	—	discharge Cd if recovery = no, else recovery coefficient	0.8–1.0 [std]
Discharge coefficient, plenum → duct	—	reverse-direction Cd	0.8–1.0 [std]
Activate reference diameter	—	yes/no	no [std]
Reference diameter	mm	shown when activated	[case]

2.4 Port (Lumbrera)

A two-stroke scavenge or exhaust window whose area varies with piston position.

Field	Unit	Meaning	Default
Height, width, top/low radius	mm	window geometry	[case]
Position with regard to BDC	mm	axial location	[case]
Reference diameter	mm	Cd reference	[case]

Field	Unit	Meaning	Default
Cd-vs-opening table	—	saved as .LUM (count, then rows of opening, inlet Cd, outlet Cd); opening is stored in metres in the file	[case]

2.5 Rotary Disk

A two-stroke crankcase intake.

Field	Unit	Meaning	Default
Reference diameter	mm	Cd reference	[case]
Cd multiplier	—	scales the tabulated Cd	1.0 [std]
Shift displacement	degree	phase offset	[case]
Speed control	—	self control / engine	engine [case]
Cd-vs-angle table	—	saved as .CD (angle in degrees, inflow Cd, outflow Cd)	[case]

2.6 Reed Valve (Lamina)

A one-way petal valve; a 0-D, 1-D, or 2-D model.

Field	Unit	Meaning	Default
Model type	—	0D / 1D / 2D	[case]
Reference diameter	mm	Cd reference	[case]
Opening direction	—	inlet-to-volume / outlet-from-volume	[case]
1-D parameters	mixed	apparent mass (g), damping, stiffness, longitudinal area (cm ²)	[case]
2-D parameters	mixed	petal density, Young's modulus, width, thickness, length, count	[case]
Cd tables	—	inlet and outlet Cd, saved as .CL; plus in/out multipliers	[case]

2.7 Wastegate Valve (mechanical)

A self-actuating spring–mass–damper wastegate (valve type 6) — the mechanical alternative to a PID-controlled wastegate. A diaphragm senses a control duct's boost and opens the valve against a spring; lift is clamped to 0–11 mm. The .WAM line is the type code 6 followed by 13 data values. The values below are the set validated on the Test5 turbo engine.

Field	Unit	Meaning	Default
Control duct	integer	duct index whose boost the diaphragm senses — the post-throttle intake plenum, not a cylinder runner	[set] 30
Distance to left end	m	sampling point along the control duct; must lie inside it	[set] 0.005
Controlling (capsule) pressure	bar	reference pressure on the diaphragm's back side	1.013 [std]
C1, C2	—	$Cd = C_1(\text{lift}/D_{\text{ref}})^{C_2}$	0.7 / 1.0 [set]
Moving-parts mass	internal	effective inertia; too large and the valve stays shut; 0 errors	5e-5 [set]

Field	Unit	Meaning	Default
Spring damping	internal	near-critical, to stop chatter	2e-3 [set]
Spring constant	internal	sets equilibrium lift; softer opens further	0.05 [set]
Spring preload	internal	holds it shut; cracking boost $\approx P_{\text{capsule}} + \text{preload} / A_{\text{diaphragm}}$	3.5e-4 [set]
Diaphragm area	m ²	boost acts here	2e-3 [set]
Plate area	m ²	exhaust/plenum pressure acts here	1.4e-3 [set]
Reference diameter	m	Cd–lift reference	0.012 [set]
Actuation type	—	“pressure in volume opens the valve” (stored 0) / “closes” (stored 1)	opens [case]

Note. The force balance runs in internal units (pressures bar, areas m², lift m), so the mechanical numbers are small values in those units, *not* the SI the labels imply — verify the saved .WAM against the set above. Figure 2 shows a wastegate converging to its steady opening.

2.8 Turbine Stator / Turbine Rotor

The nozzle and rotor stages between the exhaust and a turbine volume (Figure 3).

Field	Unit	Meaning	Default
Inlet discharge coefficient	—	stage inlet Cd, tuned to the map	0.3–0.5 [case]
Outlet discharge coefficient	—	stage outlet Cd	0.25–0.5 [case]
Reference diameter	mm	Cd reference	[case]

2.9 Node External Calculation — non-functional

A MATLAB-driven valve. **Disabled in this build; do not use it.**

3 0D Elements tab

3.1 Constant Volume Chamber (plenum)

The general-purpose plenum (Figure 4).

Field	Unit	Meaning	Default
Plenum volume	cm ³	internal volume	[case]
Initial temperature	°C	starting gas temperature	25 [std]
Initial pressure	bar	starting gas pressure	1.013 [std]

3.2 Variable Volume Chamber

A piston-driven chamber (for example, a two-stroke crankcase).

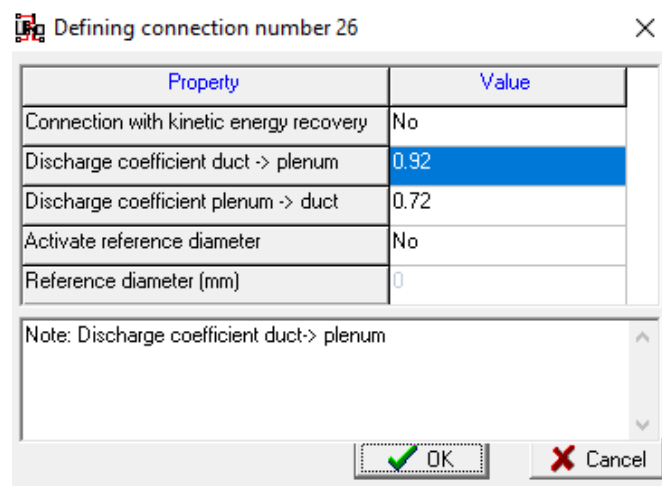


Figure 1. Node to Deposit: the connector between a pipe and a 0-D volume.

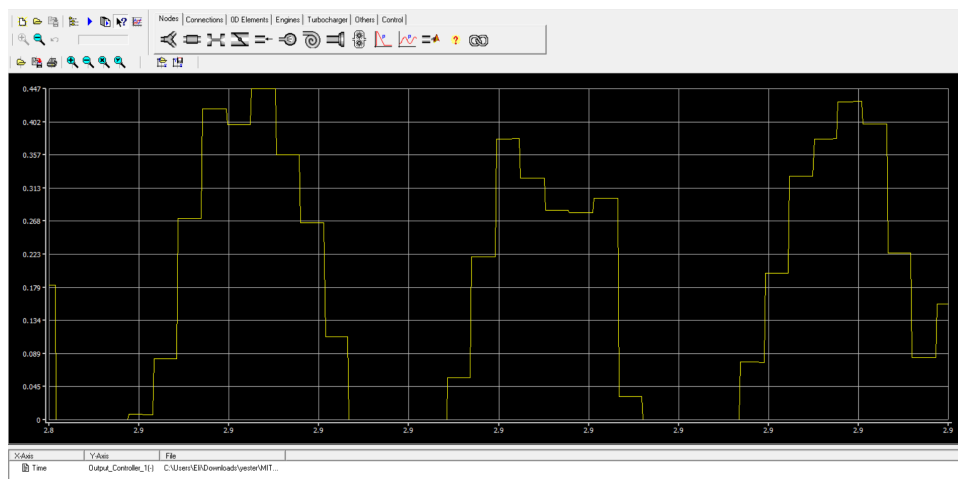


Figure 2. A wastegate's position against time (3B21T at 3250 rpm): it hunts up and down and damps to a steady opening.

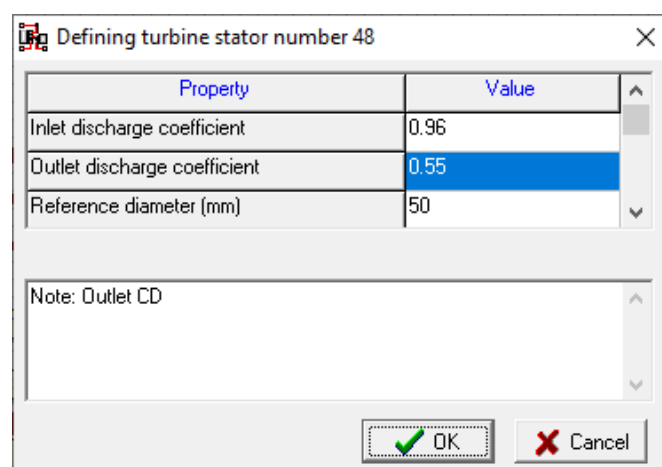


Figure 3. Turbine stator/rotor: inlet and outlet discharge coefficients and a reference diameter.

Property	Value
Plenum volume (cm3)	2500
Initial temperature (°C)	45
Initial pressure (bar)	1.95

Note: Initial temperature (°C)

Data Composition

OK Cancel

Figure 4. A constant-volume chamber: volume and initial state.

Field	Unit	Meaning	Default
Volume, initial T, initial P	cm ³ , °C, bar	as the constant-volume chamber	[case]
Conrod length, stroke, bore, misalignment	mm	kinematics of the moving wall	[case]
Compression ratio	—	volume ratio	[case]
Angular phase shift vs cylinder 1	degree	phasing	[case]
Speed control	—	engine-driven / externally driven	engine [case]
Speed ratio / Speed	—	ratio to engine (engine-driven) or fixed speed (external)	[case]

3.3 Venturi

A plenum with a throat model (deposit code 4): a main inlet, a main outlet, and a lateral (throat/suction) inlet, each through a Node to Deposit. The main flow's throat depression assists suction on the lateral port.

Field	Unit	Meaning	Default
Volume, initial T, initial P	cm ³ , °C, bar	as a plenum	[case]
Section/area ratio	—	main-inlet to throat area ratio (> 1 = contraction). Not exposed in the UI — written as 0, which makes the venturi inert; set > 0 by hand (validated at 2.0) to engage it	0 (inert) [case]
Evolution efficiency	—	throat velocity-recovery efficiency, 0–1	0.9–1.0 [std]

Field	Unit	Meaning	Default
Heat loss	—	subtracted in the throat velocity term; keep near 0 to avoid a negative root	0 [std]

3.4 Directional Junction

A merge of two inlets to one outlet with an anti-backflow rule (deposit code 5): a fast branch suppresses backflow into a slow one. Three ducts through Node to Deposit (inlet 0, inlet 1, outlet).

Field	Unit	Meaning	Default
Volume, initial T, initial P	cm ³ , °C, bar	as a plenum	[case]
Outlet Cd maximum (per inlet)	—	max reverse-flow Cd, 0–1	0.6–1.0 [case]
Cut speed (per inlet)	m/s	below the other inlet's velocity, reverse Cd stays at max	[case]
End speed (per inlet)	m/s	above it, reverse Cd is forced to 0; must differ from cut speed	[case]

Note. In steady co-flow the reverse-Cd rule is dormant; cut/end matter only under reversing flow. Set them to bracket the real operating velocity, not the tiny UI placeholders.

4 Engines tab

4.1 4T / 2T Engine (block)

The engine block carries the working conditions, geometry, heat transfer, mechanical losses, combustion, and firing order. The most consequential fields:

Field	Unit	Meaning	Default
Motor type	—	spark / compression ignition	[case]
Engine operating condition	—	steady / transient constant speed / transient variable speed / external speed control	steady [case]
EGR's calculation	—	enables burnt-gas (EGR) species tracking; does nothing physical on its own (build the loop from pipes and a valve)	no [std]
Initial engine speed	rpm	starting speed	[case]
Injected fuel mass	mg	per cylinder-cycle; compression ignition only	[case]
Relative air fuel ratio	—	this is $\varphi = F_r = 1/\lambda$, not λ ; spark ignition only	[case]
Fuel heating power / density	kJ/kg, kg/m ³	fuel properties	[case]
Duct number of volumetric efficiency	integer	the duct where breathing flow is measured for VE	[case]
Combustion model	—	Imposed heat-release law (Wiebe) / ACT (compression ignition, extended species)	Wiebe [case]

Field	Unit	Meaning	Default
CW1, CW2, Xpe	—	Woschni heat-transfer coefficients: piston-speed (≈ 2.28), swirl ($\approx 3e-3$), and combustion term ($0 \rightarrow$ default 0.001 ; DI $\approx 3.24e-3$)	see meaning
Mechanical losses K0–K3	—	friction correlation $Mecloss = K_1 \frac{N}{60} - K_2 (\frac{N}{60})^2 + K_3 \text{bmip} + K_0$	[case]
Torque to maximum power	Nm	feeds the loss correlation; must be nonzero (the UI errors on 0)	[case]

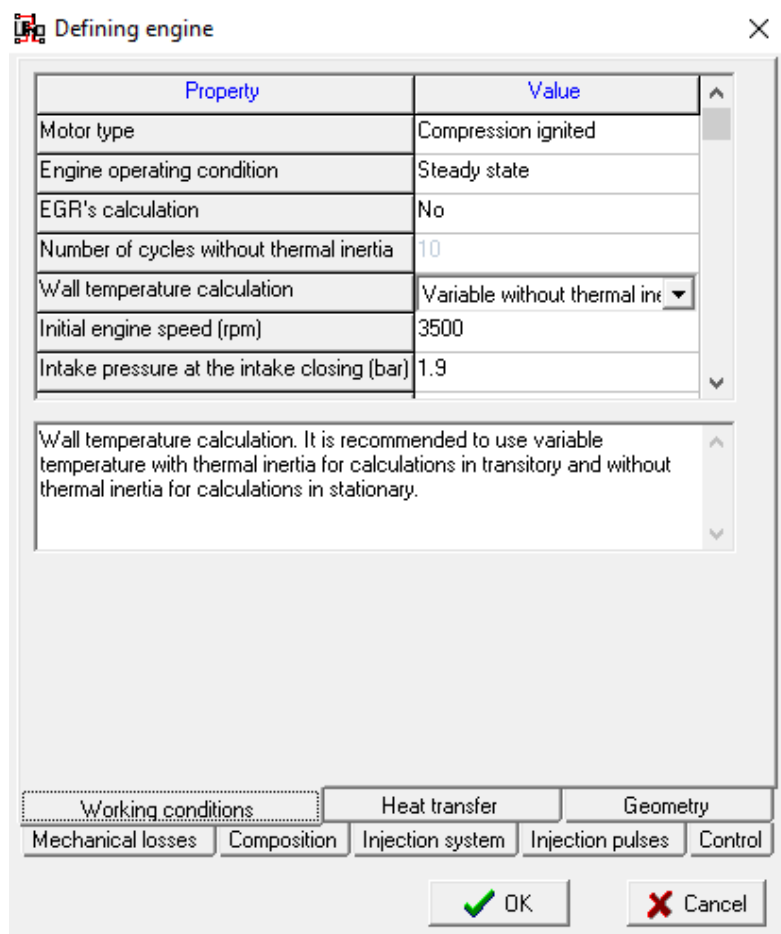


Figure 5. The engine block, showing the working-conditions, heat-transfer, geometry, mechanical-loss, and combustion tabs.

Vehicle Model (transient variable speed only)

A longitudinal vehicle model; engine speed evolves from vehicle dynamics.

Field	Unit	Meaning	Default
Rolling resistance A0/B0/C0, exponent n	N, —	rolling force = $A_0 + B_0 v + C_0 v^n$	[case]
Aero coefficient CD, air density, frontal area	—, kg/m ³ , m ²	aero force = $\frac{1}{2} C_D \rho A v^2$	[case]

Field	Unit	Meaning	Default
Road angle	degree	grade; force = $Mg \sin(\text{angle})$	0 [case]
Vehicle / transmission / wheel mass	kg	total mass = $m_v + m_t + 4m_r$	[case]
Engine / gearbox / wheel inertia	kg m ²	three inertias (the middle one is easy to miss)	[case]
Gearbox ratio / efficiency, final-drive ratio	—	drivetrain	[case]
Wheel radius	m	speed-to-rpm relation	[case]

Note. Enable Vehicle_Speed and Resistant_Torque to watch a pull. With no gear-shift model, a single-gear wide-open pull over-revs — bound the transient or balance the road load.

4.2 4-stroke / 2-stroke Cylinder

Per-cylinder geometry, combustion, and heat-release. The engine cycle is 0–720° with 0 = firing TDC and 360 = overlap TDC. The heat-release law rows carry Ma (g), Mf (mg), rpm and the Wiebe coefficients m (form exponent), C (completeness, ≈ 6.9), β (fraction of Mf burned by this sub-law; the rows sum to 1), burn duration, and Alpha0 (entered positive as ° before firing TDC; the reader negates it).

4.3 ACT (predictive diesel combustion)

A spray-packet apparent-combustion-time model; **requires the extended species model**; computes NOx. It opens two injection tabs.

Field	Unit	Meaning	Default
Number of orifices, nozzle diameter, nozzle Cd	—, m, —	injector geometry	[case]
Injection pressure	bar	rail pressure	[case]
Coefficients A1–D2	—	spray/combustion tuning	[set]
Injection pulses	—	angle-vs-mass-percent table	[case]

Note. First-entry UI bug: the injection-pulses tab divides the angle by 1000 and the mass percent by 10 on the first save. Re-open and re-type to fix; it does not recur.

5 Turbocharger tab

5.1 Turbocharger (shaft)

The shaft tying compressor and turbine (Figure 6).

Field	Unit	Meaning	Default
Speed for case	—	variable speed / fixed speed	[case]
Initial speed	rpm	starting shaft speed — set near the expected equilibrium	[case]

Field	Unit	Meaning	Default
Shaft moment of inertia	kg m ²	read only in variable-speed mode ; sets settling time, not the equilibrium	[case]

Note. In fixed-speed mode the inertia field is inert; if the UI writes an inertia token into the shaft line in fixed mode, delete it — the extra token desynchronises the parse.

5.2 Turbine

A 0-D turbine volume plus a map or a 2-nozzle model (Figure 7).

Field	Unit	Meaning	Default
Plenum volume, initial T, initial P	cm ³ , °C, bar	turbine volume state	[case]
Turbine model	—	2-Nozzle / Controlled 2-Nozzle (VGT) / Map	[case]
Wheel diameter	mm	rotor diameter	[case]
Efficiency multiplier	—	scales the efficiency	1.0 [std]
Efficiency model	—	(2-nozzle models only; hidden for Map) Watson / external / real-curve polynomial, with peak-efficiency and bound blade-speed ratios	[case]

5.3 Twin Flow Turbine

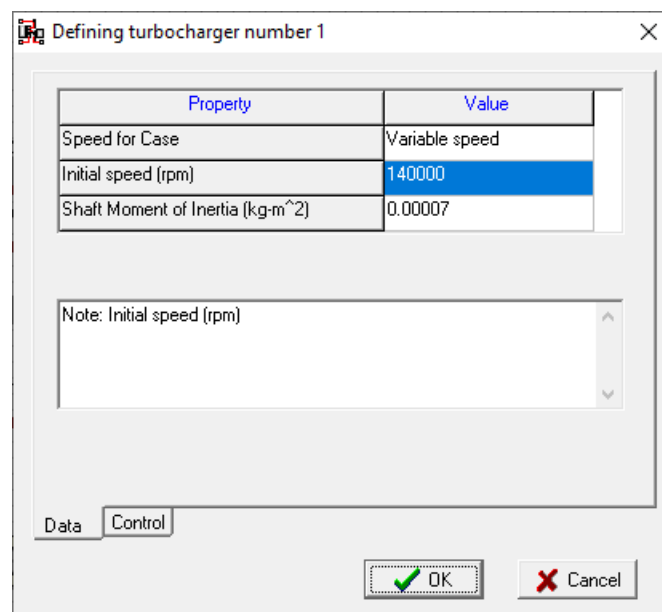
Two stators feeding one rotor from two scrolls (deposit code 3); the single map is shared, each stator taking half the effective area, for exhaust pulse separation. All fields are identical to the single turbine. Set the initial shaft speed near equilibrium.

5.4 Compressor

The intake-side compressor and its map (Figure 8). Its suction attaches to a constant-volume chamber (its inlet volume).

Field	Unit	Meaning	Default
Map input mode	—	WAMer / SAE (prefer SAE)	SAE [std]
Reference pressure / temperature	bar / °C	map reference conditions	[case]
Dampening time constant	s	response smoothing	[case]
Mass multiplier	—	scales mass flow	1.0 [std]
Corrected-speed / mass-flow ranges	rpm / g s ⁻¹	map extents (WAMer mode only)	[case]
Map	—	imported .CMP (SAE or WAMer format)	[case]

Note. Set the inlet reservoir to a real 1.013 bar and ambient temperature, never 0, or the compressor draws from vacuum and NaNs.



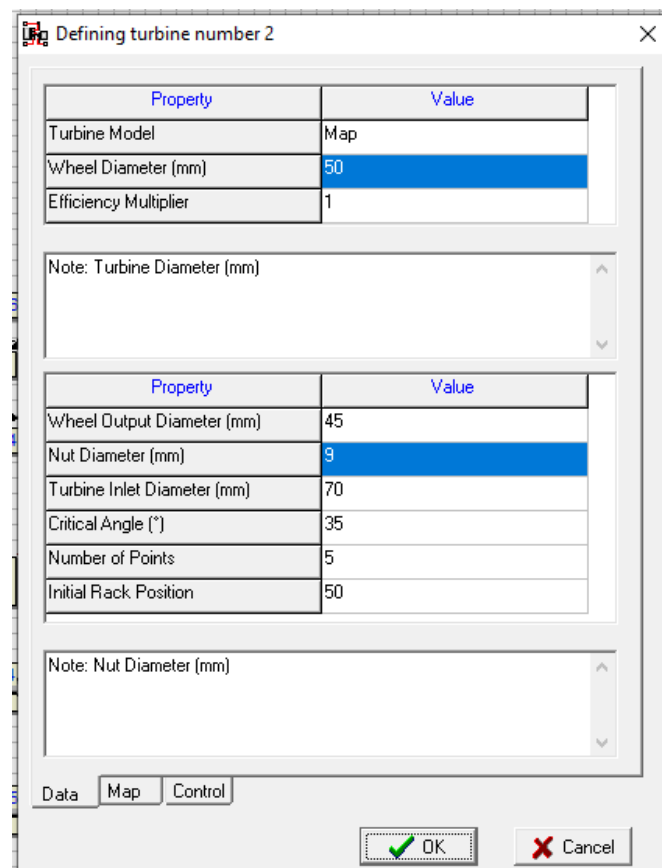
Property	Value
Speed for Case	Variable speed
Initial speed (rpm)	140000
Shaft Moment of Inertia (kg-m ²)	0.00007

Note: Initial speed (rpm)

Data Control

OK Cancel

Figure 6. The turbocharger shaft: speed mode, initial speed, and inertia.



Property	Value
Turbine Model	Map
Wheel Diameter (mm)	50
Efficiency Multiplier	1

Note: Turbine Diameter (mm)

Property	Value
Wheel Output Diameter (mm)	45
Nut Diameter (mm)	9
Turbine Inlet Diameter (mm)	70
Critical Angle (°)	35
Number of Points	5
Initial Rack Position	50

Note: Nut Diameter (mm)

Data Map Control

OK Cancel

Figure 7. The turbine: volume, model, wheel diameter, and efficiency.

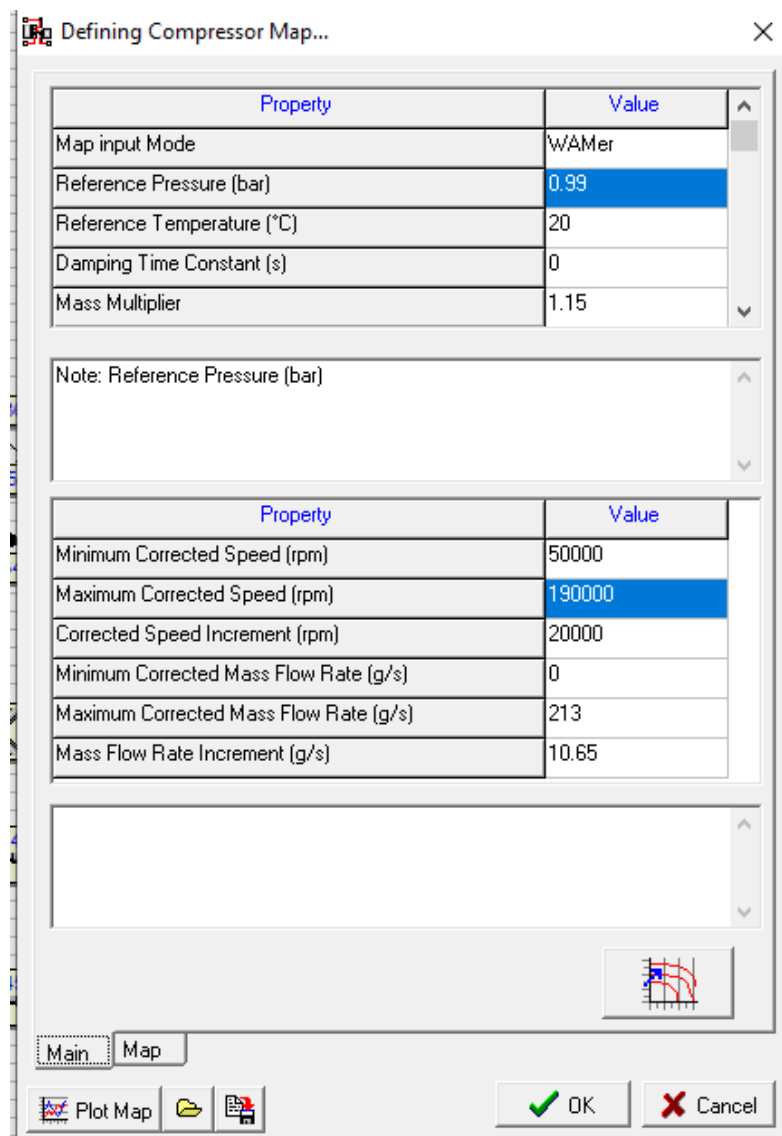


Figure 8. The compressor: map reference conditions and map import.

6 Others tab

6.1 Execution (Ejec)

The mandatory, invisible node carrying the run-wide settings.

Field	Unit	Meaning	Default
Calculation methodology	—	Common (one global timestep) / Independent (per-pipe substepping)	[case]
Duration	cycles or s	engine <i>cycles</i> for engine models, <i>seconds</i> for engine-less rigs	[case]
Specific heat ratio	—	constant / $f(Y)$ / $f(Y, T)$	$f(Y, T)$ [std]
Ambient pressure / temperature	bar / °C	atmospheric conditions	1.013 / ambient [std]
Species calculation	—	simplified (3) / extended (9); extended required for ACT	[case]
Numerical method	—	Lax-Wendroff / TVD (RoeM) / CE-SE (broken — do not use)	TVD [std]
Courant number	—	stability number; lower for hot exhaust	0.8 [std]
AngleStep / Plot timestep	°CA or s	instantaneous-output sampling interval (seconds when engine-less)	[case]
Results output mode	—	Last Cycle / every cycle split / concatenated / every Nth cycle	Last Cycle [std]
Space-time magnitudes	—	Pressure/Temperature/Velocity/Mass-flow/Species selection for wave maps	none [case]

7 Control tab

Figure 9 shows a live control chain: the green logic lines link sensor, 1D table, and PID, while the red “C” marks the controlled wastegate (which has no line to it). The wiring rules are in the *User Guide*, control chapter.

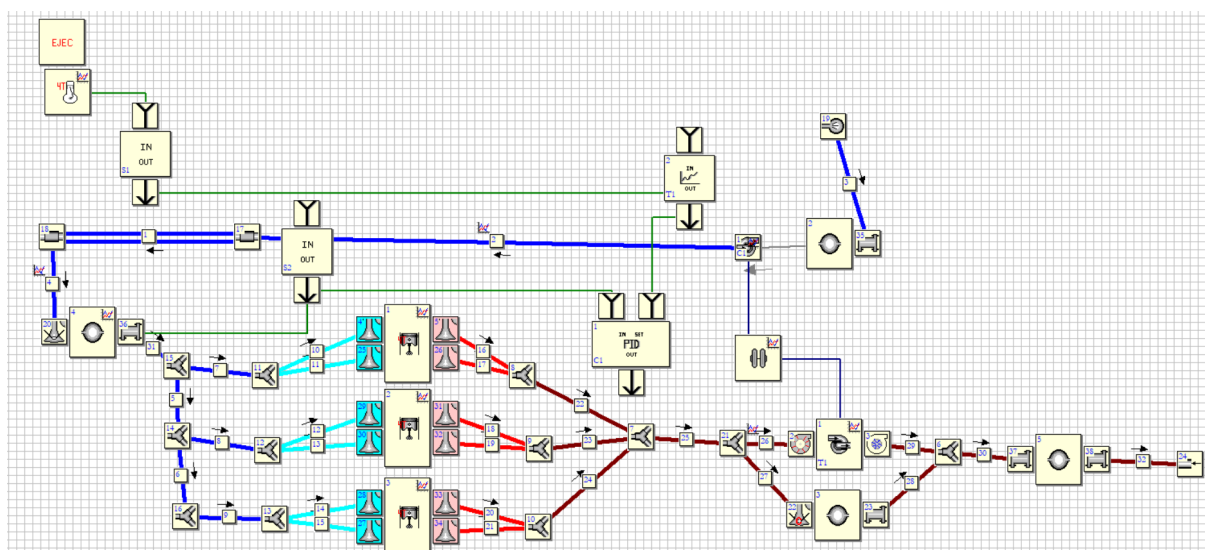


Figure 9. A boost-control chain in context: green logic lines linking sensor → 1D table → PID, and the red “C” on the PID-controlled wastegate.

7.1 Sensor (In/Out)

Field	Unit	Meaning	Default
Input data	—	the measured quantity, auto-populated from the connected component	[case]
Time constant	s	measurement lag	[case]
Gain	—	output scaling; 0 silently disables the sensor	1.0 [std]
Sensor position	m	for position-meaningful sensors (e.g. along a duct)	[case]

Note. The sensor index is 1-based — a subtle off-by-one trap when wiring by index.

7.2 PID

Field	Unit	Meaning	Default
P/I/D positive, P/I/D negative	—	separate gains for positive and negative error	[case]
Initial output / Constant output	—	starting value; a fixed output if used open-loop	[case]
Maximum / Minimum output	—	saturation limits (anti-windup holds the integral at the limit)	[case]
Period	—	sample period	[case]
Time constant / Gain	—	output filtering and scaling	[case]
Setpoint	—	target value	[case]

Note. The derivative differentiates every solver step, but a slower sensor feeds it a piecewise-constant signal, so it contributes little in practice — tune with P and I first.

7.3 Table 1D (In/Out interpolation)

Field	Unit	Meaning	Default
Period	—	refresh period	[case]
Interpolation mode	—	Linear / Hermite / Steps	Linear [case]
Read data from file	—	yes/no; when yes, a .T1D filename	no [case]
Table data	—	$X \rightarrow Y$ points, inline or from .T1D (two columns, no header/count)	[case]

7.4 Switch / Decisor

A logic selector controller (fields: target, high/low thresholds). Used to switch a driven value on a condition.

7.5 Gain

A scalar multiplier block: one gain field.

8 Features (not palette components)

8.1 Duct

The connector, with its own physics.

Field	Unit	Meaning	Default
Spatial mesh	mm	cell size	[case]
Duct type	—	intake pipe / exhaust pipe / exhaust port / intake port (chooses the wall heat-transfer correlation; <i>pipe</i> shows the Heat Transfer tab, <i>port</i> pins the wall to engine coolant)	[case]
Wall temperature calculation	—	constant / variable with (or without) thermal inertia	constant [std]
Duct roughness	mm	stored in mm , an exception to the SI-in-file rule	[case]
Initial wall / gas T, gas P	°C, bar	starting state	[case]
Initial gas velocity	m/s	0 is fine on TVD/RoeM	0 [std]
Heat-transfer / friction correction	—	multipliers	1.0 [std]
N of intercooler ducts	integer	repeats the duct <i>N</i> times to mimic an intercooler	1 [std]
Inlet / outlet diameter, length	mm	geometry; area variation is stable on TVD/RoeM	[case]

Note. The “Rigid” right-click toggle is cosmetic canvas routing only — it is not written to the .WAM and does not affect the solver.

8.2 EGR flag

The engine working-conditions yes/no flag that adds a burnt-gas marker species. It carries no physics on its own; build the recirculation loop from ordinary pipes and a valve (User Guide, engine chapter).

8.3 Space-time results

Per-magnitude $x-t$ wave maps. Requires both the element checkbox and the Execution-node magnitude selection; output is one tab-separated, labelled file per magnitude for the last cycle. Pick pipes (a pipe gives one column per cell).

8.4 Parametric study

A WAMer batch-sweep generator: it writes one .WAM per case plus a .gro batch. It is not part of the .WAM and the solver has no concept of it. Manual case entry works; the .DAT/curve auto-import softlocks WAMer — do not use it. Do not sweep a model containing a mechanical wastegate (its mass field is zeroed on regeneration; now guarded).

8.5 Heat transfer / wall temperature

Constant (default) or variable with/without thermal inertia; coolant is air or water; wall layers support an intercooler model. Keep constant unless modelling an intercooler.

9 File formats

Extension	Format
.WAM	the model (text), read by the solver
.PCS	WAMer project (binary), editor-only
.DAT	results, tab-separated with a labelled header row
.LEV	valve lift: line 1 = count, then one lift (mm) per line
.FLU	valve flow: line 1 = count, then rows of inflow Cd, outflow Cd, swirl
.CMP / .TMP	compressor / turbine map (SAE or WAMer format)
.CD / .CL	rotary-disk / reed-valve discharge coefficients
.LUM	port discharge coefficients (opening stored in metres)
.T1D	1D-table data: two whitespace-separated columns, X then Y , no header
.gro	batch list: line 1 = count, then three lines per run (exe, input WAM, output base)
.AXI / .GRW / .PLT	plot axes / saved graph / results-file template

A sample of each ships in the repository.

10 Results magnitudes — the selectable catalogue

Every quantity the per-element results tabs can enable. **AVG** = a per-cycle scalar (to AVG.DAT); **INS** = a per-step trace over the last cycle (to INS.DAT). Units are appended automatically (Nm, bar, kW, and so on; (kg/cc) = kg per cylinder-cycle).

10.1 Engine block — performance scalars (AVG)

The torque family — cycle-effective versus mechanism:

Label	Meaning
Net_Torque	net indicated torque (net work = gross indicated – pumping), before friction
Torque_of_mechanical_losses	the friction (FMEP) torque lost in the mechanism
Effective_Torque	brake torque = Net_Torque – Torque_of_mechanical_losses (running value)
Cicle_Effective_Torque	cycle-averaged brake torque — the steady dyno number; use this for a torque curve
Resistant_Torque	external load torque applied to the shaft (transient/vehicle runs)

Mean effective pressures, work, and power:

Label	Meaning
IMEP_Cycle	gross indicated MEP (UI label reads “INIP”)
NMEP / NMEP_Cycle	net indicated MEP
PMEP_Cycle	pumping MEP

Label	Meaning
BMEP / BMEP_Cycle	brake MEP (the effective-torque analogue)
Net_Work / Pumping_Work	per-cycle work (J)
Power / Power_Cycle	running / cycle-averaged brake power (kW)

Breathing, efficiency, and state:

Label	Meaning
Intake_Mass / Fuel_Mass / Trapped_Mass	per cylinder-cycle masses
Volumetric_Efficiency	vs the reference-tube gas density (hot; can read > 1)
Volumetric_Efficiency_Atm	vs standard atmospheric density (the textbook VE)
Effective_Efficiency / Indicated_Efficiency	brake / indicated thermal efficiency
Break_Specific_Fuel_Consumption	BSFC (g/kWh)
Fuel_To_Air_Ratio	$F_r = \varphi$, the relative fuel/air (not λ)
AFR / Engine_Speed / Swirl	air-fuel ratio, speed (rpm), swirl
Vehicle_Speed	km/h (vehicle model only)

10.2 Cylinder — per-cycle averages (AVG)

Label	Meaning
Net_Work_Cyl / NMEP_Cyl / Pumping_Work_Cyl / PMEP_Cyl / IMEP_Cyl	work and MEPs
Combustion_Heat_Cyl	heat released (J)
Heat_Through_Liner/Cylinder_- Head/Piston_Cyl	wall heat split
Trapped_Mass_Cyl / Intake_Mass_Cyl / Exhaust_Mass_Cyl	per-cycle masses
Short_Circuit_Mass_Cyl	fresh charge lost straight to exhaust (2-stroke / overlap)
BlowBy_Mass_Cyl / Mean_AFR_Cyl / Average_Temperature_Cyl / Average_Swirl_Cyl	blow-by, AFR, mean T, swirl
Volumetric_Efficiency_Cyl	per-cylinder VE
Temp_Liner/Piston/CylHead_- Int/Med/Ext_Cyl	wall-node temperatures (inner/mid/outer)
Mean_NIT_Cyl / Mean_NIT_ValvExh	mean flux/enthalpy metric

10.3 Cylinder — instantaneous (INS)

Label	Meaning
Pressure_Cyl / Temperature_Cyl / Mass_Cyl / Volume_Cyl	in-cylinder state
Fuel_Mass_Cyl	injected fuel (mg)
HRL_Cyl	heat-release-law fraction, $0 \rightarrow 1$ (combustion phasing)

Label	Meaning
Torque_Cyl	instantaneous per-cylinder torque
Woshni_Coef_Cyl / Gamma_Cyl	Woschni HTC and γ
Y_<species>_Cyl	species mass fractions
Mass_Flow_Int/Exh_Valve, Total_Int/Exh_Mass_Flow_Cyl	valve flows
Mach_Valv_Int/Exh / Effective_Section_Valv_Int/Exh	valve Mach and open area
Angular_Momentum_* / Short_Circuit_Flow_Cyl / BlowBy_Cyl	swirl build-up, short-circuit, blow-by

10.4 Duct / pipe

Label	Meaning
P_duct / V_duct / T_duct / F_duct	pressure (bar), velocity, temperature, mass flow
V_Right/Left_duct, P_to_Right/Left_duct	Riemann wave split
NIT_duct / Internal_Heat_Trans_Coef_duct	flux metric, internal HTC
WT0/WT1/WT2_duct	wall temperatures (inner/mid/outer)
Specific_Heats_Ratio / Y_<species>	γ and species

Note. Sample duct outputs at $x = 0.01$, not exactly 0, to avoid the boundary-node NaN under a strong transient.

10.5 Plenum / chamber (0-D)

Label	Meaning
Pressure_plenum / Temperature_plenum / Volume_plenum / Mass_plenum	state
Specific_Heat_ratio_plenum / Y_<species>	γ and species

10.6 Turbomachinery

Label	Meaning
Power_comp / Efficiency_comp / Comp_Ratio_comp	compressor power, efficiency, pressure ratio
Mass_Flow_comp / Corrected_Mass_Flow_comp / Corrected_Speed_comp	compressor flow and corrected quantities
Work_turb / Efficiency_turb / Power_turb	turbine work, efficiency, power

Label	Meaning
Blade_speed_ratio / Corrected_air_mass_flow / Corrected_speed / Expansion_ratio	turbine operating point
Speed_Turbocharger_Shaft / Mechanical_power_turbo / Mechanical_efficiency_turbo	shaft speed (rpm), power, efficiency

10.7 Venturi, nodes, valves, control

Label	Meaning
Pre_entry_vent / Pre_throat_vent / Mach_entry/throat_vent	venturi pressures and throat Mach
Inflow_Discharge_Coeff_Node / Outflow_Discharge_Coeff_Node	node discharge coefficients
Lift_Wate-Gate / Lift_Sheet	wastegate lift / reed-valve lift
Output_Sensor / Input_Sensor	sensor signals
Output_Controller / Filt_Output_Controller / Error_PID	controller output and error
P_Action_PID / I_Action_PID / D_Action_PID	the three PID term contributions

10.8 Axes (always available)

Label	Meaning
Time(s) / Angle(deg) / Cycle(-)	the independent axes for any trace

10.9 Diesel particulate filter (only if compiled with the DPF option)

Label	Meaning
Soot_mass / Soot_thickness / Filtration_efficiency / Porosity	loading state
k_wall / k_soot	wall and soot permeabilities
T_wall_IC/middle/OC / Qreg / Kreg_thermal / Kreg_cat	wall temperatures and regeneration